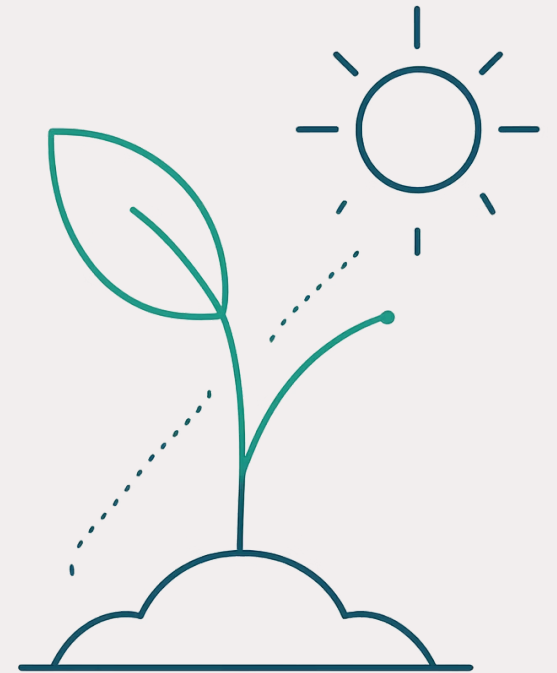


Year 6 Science Mission: Seeds of Science

National Science Week 2026 · Aussie Primary Academy

Could you solve a real-world problem using science?



What is National Science Week?

A National Celebration

Scientists, students, and communities across Australia come together to explore discoveries and share the joy of science.

This Year's Theme

"Seeds of Science: nurturing science for all" — every question you ask plants a seed of discovery.

How Science Really Works

Scientists don't memorise all the answers. They ask questions, test ideas, collect evidence, and revise their thinking when new data appears.

Your Role This Week

You're a scientist. You'll ask questions, design tests, and use evidence to solve real problems.



The Scientific Inquiry Cycle

Every scientist follows this path — from a question to a conclusion:

Question

Hypothesis



Fair Test

Variables

Quick Example: Plant Growth

- **Independent variable (what we change):** Light — sunlight vs shade
- **Dependent variable (what we measure):** Plant height in centimetres
- **Controlled variables (what stays the same):** Water, soil type, pot size, temperature

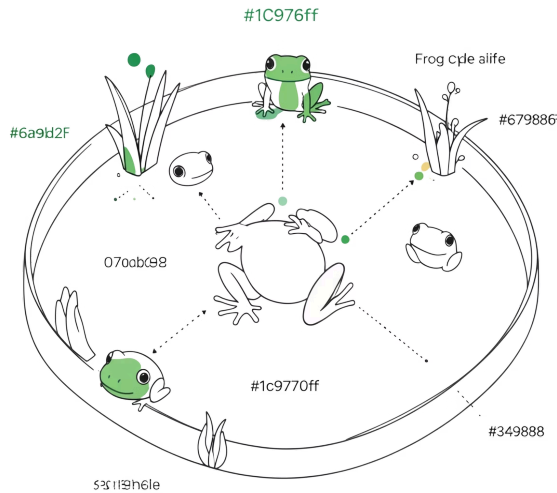
🤔 Your Turn

In the plant growth example above — **which variable is the independent variable?**

Hint: It's the one the scientist deliberately changes to see what happens.

- ❏ Science is not about memorising facts — it's about knowing how to find answers.

Life Cycles & Survival



- **All living things have life cycles:** Birth → Growth → Reproduction → Death. Each stage has different needs.
- **Environmental conditions matter:** Temperature, water, light, soil nutrients, and space all affect survival.
- **Adaptation:** Over time, species develop features that help them survive — thick fur for cold climates, long roots for dry soil.

Comparison Challenge

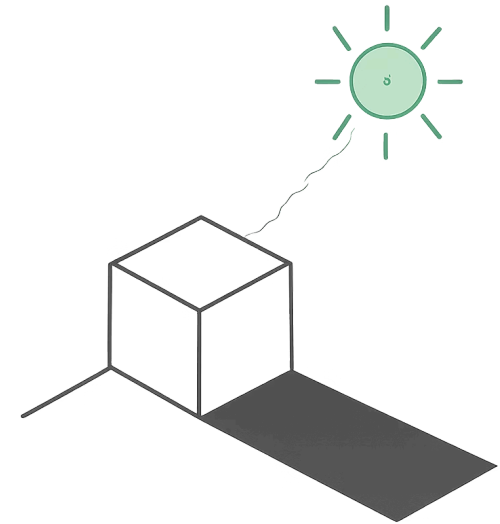
A frog tadpole needs water to survive. An adult frog can live on land. What environmental change occurs during the tadpole's life cycle that allows this?

Real-world question: How could a drought, flood, or temperature rise affect a species' ability to survive?

Earth, Sun & Shadows

How Shadows Change Throughout the Day

Why? Light travels in straight lines. The angle of sunlight determines shadow direction and length.



🤔 Reasoning Challenge

You measure a shadow at 9 am and find it is 2 metres long. At 12 pm, the same object's shadow is only 0.5 metres long.

Explain why the shadow became shorter. What evidence would you collect to prove your explanation?

Morning (9 am)

Sun is low in the east → shadows are **long** and point west.

Midday (12 pm)

Sun is high in the sky → shadows are **short** and point roughly south.

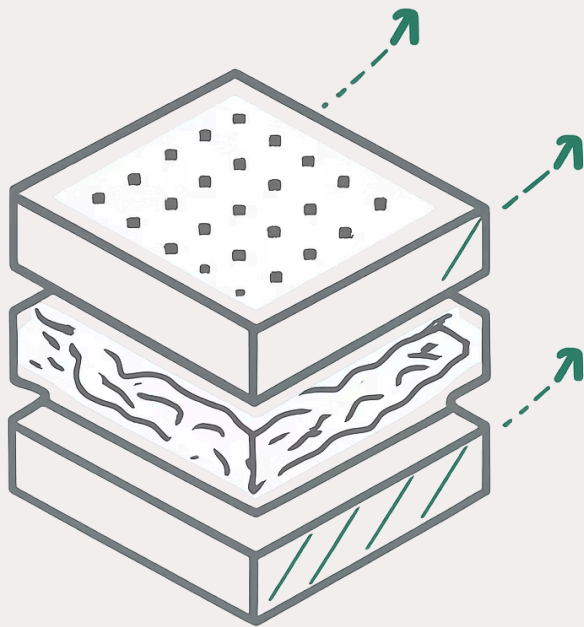
Afternoon (3 pm)

Sun is low in the west → shadows are **long** and point east.

Heat, Energy & Materials

Conductors vs Insulators

Why it matters: Choosing the right material solves real problems — keeping food hot or cold, protecting buildings, designing protective clothing.



Real-World Problem

A school wants to keep drinking water cool during a hot day. The water is stored in a **metal container**.

Which material should the school wrap around the container — **aluminium foil** or **foam insulation**?

Why? What evidence would prove your choice works?

⚡ Conductors

Allow heat to pass through quickly. Examples: aluminium, copper, steel.

🔒 Insulators

Slow down heat transfer. Examples: foam, wool, air, plastic.

Science in the Real World

Scientists solve problems in five big areas:



Environments

How do we protect habitats? How do species adapt to climate change?



Energy

How do we generate power sustainably? How do we reduce energy waste?

MDB

Materials

How do we design stronger, lighter, or more recyclable materials?



Living Things

How do we improve health, grow food, or understand disease?



Technology

How do we design tools and systems that solve human problems?

 **STEM Challenge:** A farmer's crops are dying because the soil is too dry. Using science, design a solution. What do plants need? How could you deliver water efficiently? How would you test whether your solution works?

Mini Investigation: Which Material Provides the Best Insulation?

Research Question

Which material best slows down heat loss from warm water?

Example Hypothesis

"Foam insulation will keep water warmer longer than aluminium foil because foam traps air, which is a poor conductor of heat."

Variables

Independent Variable

Type of material: foam, wool, plastic, aluminium foil, paper

Dependent Variable

Temperature of water over time (°C)

Controlled Variables

Starting temperature, container size, room temperature, time intervals, material thickness, amount of water

Fair-Test Procedure

1. Wrap each container with a different material (same thickness)
2. Fill each with hot water at the same starting temperature
3. Record temperature every 5 minutes for 30 minutes
4. Keep all containers in the same room at the same temperature
5. Repeat the test 3 times to check for reliability

Data Table to Complete

Time (min)	Foam (°C)	Wool (°C)	Foil (°C)	Paper (°C)
0				
10				
20				
30				

Evidence-Based Conclusion

"The data shows that [material] was the best insulator because [explain using results]. This supports/does not support my hypothesis because [explain]."

 **Safety:** Teacher handles hot water. Students use thermometers carefully. Wash hands after handling materials.

Science Week Challenge: Higher-Order Questions

Answer these questions using reasoning and evidence:

1 Interpret Information

A scientist tests three plant varieties in a drought. Variety A wilts after 5 days, Variety B after 10 days, Variety C after 15 days. Which variety would be best to plant in a severe drought? Why?

2 Identify Variables

A student investigates whether salt water freezes at a lower temperature than fresh water. List the independent variable, dependent variable, and three controlled variables.

3 Evaluate Evidence

Student A uses a thermometer in direct sunlight to test heat absorption by colour. Student B uses a thermometer in shade. Whose results will be more reliable? Why?

4 Explain Cause and Effect

When a forest is cleared for farming, the soil often erodes. Explain why removing trees causes soil erosion. What evidence would you collect to prove this?

5 Justify a Conclusion

A company claims its new insulating material keeps food cold for 24 hours. Design a fair test to check this claim. What data would you collect? How would you decide if the claim is true?

Teacher Answers Available Separately

Your Science Mission

Find a problem around you that could be investigated or improved using science.

01

The Problem

What is something in your school, home, or community that could be better? (e.g. "The playground gets too hot in summer," "Food spoils quickly," "We waste too much water")

03

Your Hypothesis

What do you think will happen and why?

05

Evidence to Collect

What data would you gather? How would you measure it?

02

Your Question

What specific question would you ask? (e.g. "Which surface material stays coolest in the sun?")

04

What You Would Test

What variables would you change and measure?

06

How You'd Know It Worked

What would the data need to show to prove your idea works?

"Scientists don't just know answers. They know how to find them."

Your curiosity, your questions, and your willingness to test ideas — that's what makes you a scientist. 